MOHSIN ZAHID

Senior Agentic AI Engineer | Multi-Agent Systems & MCP Tooling

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Senior agentic AI engineer and backend/platform builder focused on production multi-agent systems, MCP tooling, and secure workflow automation. Owns systems end to end across NestJS/Node.js and Python: agent routing, delegated execution, approvals, memory and context isolation, multi-tenant boundaries, integrations, observability, and product delivery. Builds AI-native developer workflows that research, plan, implement, verify, and report with explicit human control and failure handling.

CORE COMPETENCIES

Agentic AI: Multi-agent orchestration, agent fleets, routing/delegation, autonomous software engineering, bounded autonomy, approval gates, memory/context isolation, retries/fallbacks, audit trails, cost/action limits.

MCP & integrations: MCP client/server tooling, discovery and entitlements, project-scoped sharing, private OAuth, secure connector handling, Slack, GitHub, Jira, Telegram, Google, Notion, webhooks.

Backend & platform: NestJS, Node.js, TypeScript, Python, FastAPI, REST APIs, PostgreSQL, SQLite, JWT, Pydantic, SQLAlchemy, multi-tenant architecture, state isolation.

Product & frontend: Next.js, React, React Native, Lit, Vite, Streamlit, workflow builders, operational dashboards, evidence viewers, accessibility and validation.

Infrastructure & engineering: Docker, Docker Compose, AWS, Cloudflare R2, Linux/VPS, CI/CD, monitoring, observability, TDD, code review, security review, QA, documentation.

PROFESSIONAL EXPERIENCE

Backend Lead / Agentic AI Engineer — MUST Company

Jun 2025 – Present · Remote, Pakistan

- Built and advanced Pulse Workspace, a multiplayer Slack/web agent workspace with isolated projects, shared context, files, connected tools, approvals, memory, skills, scheduled jobs, and secure workflow execution.
- Designed the Pulse MCP Bridge for processed, project-scoped context with explicit security and permission boundaries; delivered rich AI-tool interfaces inside workspace chat.
- Delivered member self-service MCP and OAuth capabilities: external MCP registration, project-scoped sharing, private OAuth clients, connector selection, and entitlement handling.
- Architected workflows covering custom agent creation, Slack invocation, run cards, drafts, approvals, effects, audit, recovery, ownership, lifecycle controls, and Meeting Secretary / PM Operations templates.
- Built workflow-builder and operator-scan experiences with receipts, stage timelines, execution visibility, and MCP selection; supported multi-tenant routing, onboarding, and state isolation.
- Built and operated Hermes autonomous developer workflows using Kanban dispatch and worker → verifier → synthesizer swarms, isolated workspaces, human review, model routing, retries, fallbacks, and auditability.
- Contributed to HR onboarding voice-agent workflows, automated document organization, structural permissions, evidence-oriented operations, rollout, QA, documentation, and stakeholder enablement.

Full Stack Engineer — MUST Company

Dec 2022 – May 2025 · Remote

- Built and maintained production fintech SaaS systems across backend and frontend, owning API design, authentication, data architecture, and feature delivery.
- Worked across TypeScript/Node.js, Python, React, data layers, integrations, and production troubleshooting.

React Native Developer — Instamobile.io

Feb 2022 – Feb 2023 · Remote, Bucharest

- Developed cross-platform mobile applications for client SaaS products.

React Native Developer — Upwork (Freelance)

Nov 2020 – Nov 2021 · Remote

- Delivered mobile applications for multiple independent clients.

Web Developer — mooveat.co

Jan 2019 – Apr 2019 · Remote

Web Developer — Zahid Scan

Aug 2017 – Dec 2018 · Lahore, Pakistan

SELECTED AGENTIC AI & MCP PROJECTS

- **Pulse Workspace:** Production AI teammate for Slack/web teams with project-scoped context, connected tools, approvals, memory, scheduled workflows, GitHub/Notion integrations, and fleet handoff. pulse.must.company
- **dev-os:** Created an autonomous software-engineering orchestrator that researches, plans, passes one approval gate, dispatches implementation to a build team, and reports results. GitHub
- **hermes-devcrew:** Built a nine-agent engineering fleet spanning architecture, backend, frontend, DevOps, review, integration, domain expertise, QA, and design, coordinated through Kanban swarm/dispatcher mechanisms. GitHub

- **CodeRover / @coderover/mcp:** Published an MCP integration for graph-native code intelligence: cited codebase answers, autonomous PR review, security patrol, and dead-code refactoring across 16 languages using Tree-sitter and Memgraph. coderover.must.company
- **MatchHire MCP & recruitment agents:** Worked on recruiter/employer workflows, sourcing, candidate import, pipelines, matching, LinkedIn/Drive/contacts integrations, AI interviews, and Pulse integration.
- **MUST Agents / DevOS Control Plane:** Agent profiles, distribution packages, deployment workflows, fleet routing, state management, and operational observability. GitHub

ADDITIONAL OPEN SOURCE

- **NousResearch/hermes-agent:** Upstream contributions for reviewer/QA policy controls and merge-aware profile installation/update behavior, including preservation of user skills and cron jobs; PRs #67954 and #44386.
- **codeatlas:** Searchable, queryable, IDE-native source-code knowledge base. GitHub
- **qm / devcrew-bridge / control-plane:** Multiplayer agent harness, runtime bridge, and agent control-plane/observability work. GitHub
- **Miraj Dashboard:** Next.js/FastAPI/ccxt crypto dashboard with PostgreSQL, scheduled analysis, confluence scoring, and Telegram/Discord alerts. GitHub

EDUCATION

Bachelor of Science in Computer Science — University of Central Punjab, Pakistan · 2016 – 2020

LANGUAGES

English (professional working proficiency) · Urdu (native)

SELECTED TECHNICAL STACK

NestJS · Node.js · TypeScript · Python · FastAPI · PostgreSQL · SQLite · React · Next.js · React Native · Docker · AWS · Cloudflare R2 · Slack · GitHub · Jira · Telegram · Google · Notion · MCP · Claude Code · OpenRouter · Anthropic · OpenAI · Grok · Codex